

413/613 Fall 2026 - Applied Statistics & Data Science I - Lecture 3

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RECALL

From counting outcomes to random variables to distributions

$$\text{experiment} \longrightarrow \Omega \longrightarrow X(\omega) \longrightarrow P(X = x).$$

For N independent Bernoulli trials,

$$H \sim \text{Binomial}(N, p), \quad P(H = h) = \binom{N}{h} p^h (1-p)^{N-h}.$$

$$\mathbb{E}[H] = Np, \quad \text{Var}(H) = Np(1-p).$$

In this lecture, we ask how this distribution behaves in different regimes and how other important probability distributions emerge from it.

1 Cumulative Distribution Function

For any random variable X , the **cumulative distribution function (CDF)** is defined as

$$F_X(x) = P(X \leq x).$$

The CDF gives the probability accumulated up to the value x . It satisfies

$$0 \leq F_X(x) \leq 1, \quad F_X(x) \text{ is nondecreasing,}$$

with

$$\lim_{x \rightarrow -\infty} F_X(x) = 0, \quad \lim_{x \rightarrow \infty} F_X(x) = 1.$$

If X is discrete,

$$F_X(x) = \sum_{k \leq x} P(X = k).$$

Thus, the CDF is a **step function**, with jumps at the possible values of X (Fig. 1).

If X has probability density $p(x)$,

$$F_X(x) = \int_{-\infty}^x p(y) dy.$$

Hence,

$$P(a < X \leq b) = F_X(b) - F_X(a).$$

For a Gaussian random variable

$$X \sim \mathcal{N}(\mu, \sigma^2),$$

the CDF is

$$F_X(x) = \int_{-\infty}^x \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left[-\frac{(y-\mu)^2}{2\sigma^2}\right] dy.$$

The Gaussian density is bell-shaped, while its CDF is a smooth, S-shaped curve increasing from 0 to 1 (Fig. 2).

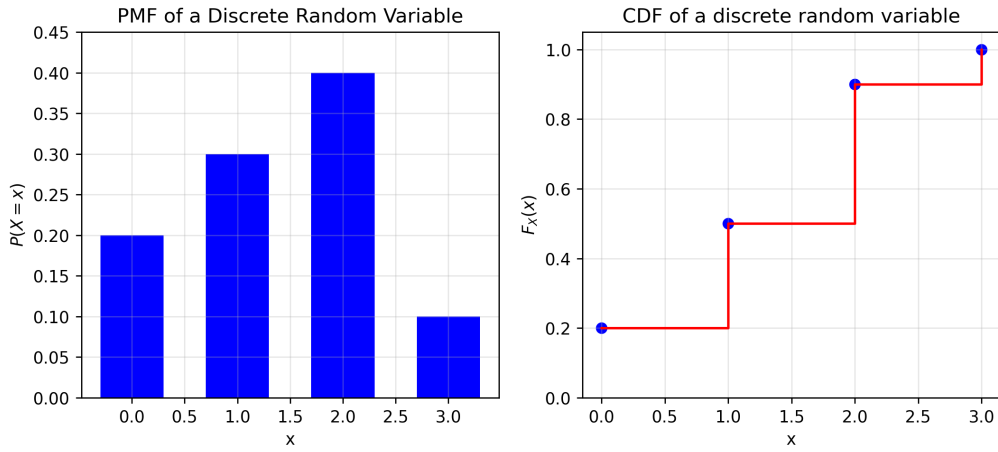


Figure 1: PMF and CDF of a discrete random variable X taking values $x \in \{0, 1, 2, 3\}$ with probability mass function $P(X = 0) = 0.2, P(X = 1) = 0.3, P(X = 2) = 0.4, P(X = 3) = 0.1$ is a step function, and the jump at each value x has size $P(X = x)$.

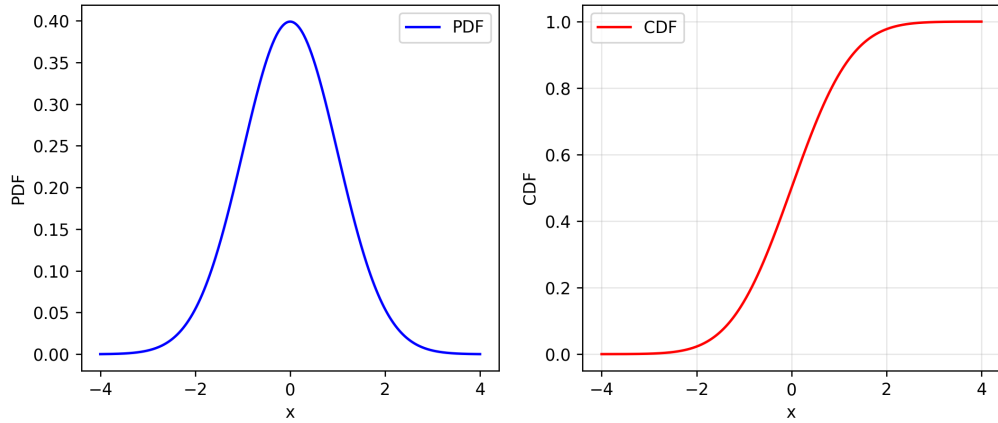


Figure 2: Probability density function (PDF) and cumulative distribution function (CDF) of a Gaussian random variable $X \sim \mathcal{N}(\mu = 0, \sigma^2 = 1)$. The CDF therefore represents the accumulated area under the Gaussian density up to x .

2 From the Binomial Distribution to Other Probability Models

Recall that if H denotes the number of successes in N independent Bernoulli trials, each with success probability p , then

$$H \sim \text{Binomial}(N, p), \quad P(H = h) = \binom{N}{h} p^h (1-p)^{N-h},$$

for $h = 0, 1, \dots, N$. The mean and variance are

$$\mathbb{E}[H] = Np, \quad \text{Var}(H) = Np(1-p).$$

The binomial distribution is an exact finite- N model. However, different limiting regimes of the same model lead to other important and commonly occurring probability distributions.

2.1 Large N : The Gaussian Approximation

The binomial distribution has mean

$$\mu = Np$$

and variance

$$\sigma^2 = Np(1-p).$$

When N is large and p is not too close to 0 or 1, the binomial probability mass function (pmf) is approximated by the Gaussian density evaluated near h ,

$$p(h) \approx \frac{1}{\sqrt{2\pi Np(1-p)}} \exp \left[-\frac{(h - Np)^2}{2Np(1-p)} \right].$$

Equivalently,

$$H \approx \mathcal{N}(Np, Np(1-p))$$

The approximation is accurate when both $Np \gg 1, N(1-p) \gg 1$. By centering the random variable H at its mean $\mu = Np$ and scaling by its standard deviation $\sigma = \sqrt{Np(1-p)}$, we obtain the standard normal distribution

$$Z = \frac{H - Np}{\sqrt{Np(1-p)}} \approx \mathcal{N}(0, 1), \quad \text{for large } N.$$

Why does a Gaussian appear? The binomial random variable can be written as a sum

$$H = X_1 + \cdots + X_N,$$

where

$$X_i \sim \text{Bernoulli}(p)$$

are independent. Since

$$\mathbb{E}[X_i] = p, \quad \text{Var}(X_i) = p(1-p),$$

we have

$$\mathbb{E}[H] = Np, \quad \text{Var}(H) = Np(1-p).$$

The fact that a properly centered and rescaled sum of many independent random variables approaches a Gaussian distribution is the content of the **Central Limit Theorem**, which we will study more carefully later.

2.2 Rare Events: The Poisson Limit

A different limiting regime occurs when an individual event is rare, i.e. $p \rightarrow 0, N \rightarrow \infty$ with the new variable $\lambda = Np$ being fixed. The binomial distribution can be rewritten now as a function of the new variable λ as follows:

$$P(H = h) = \binom{N}{h} \left(\frac{\lambda}{N} \right)^h \left(1 - \frac{\lambda}{N} \right)^{N-h}.$$

For fixed h and large N ,

$$\binom{N}{h} = \frac{N(N-1) \cdots (N-h+1)}{h!} \approx \frac{N^h}{h!},$$

and

$$\left(1 - \frac{\lambda}{N} \right)^N \rightarrow e^{-\lambda}.$$

The last simplification comes from the identity $\lim_{N \rightarrow \infty} \left(1 + \frac{x}{N} \right)^N = e^x$. Also,

$$\left(1 - \frac{\lambda}{N} \right)^{-h} \rightarrow 1.$$

Therefore,

$$P(H = h) \rightarrow \frac{\lambda^h}{h!} e^{-\lambda}.$$

This defines the **Poisson distribution**:

$$H \sim \text{Poisson}(\lambda), \quad P(H = h) = e^{-\lambda} \frac{\lambda^h}{h!}, \quad h = 0, 1, 2, \dots$$

Mean and variance. For a Poisson random variable,

$$\boxed{\mathbb{E}[H] = \lambda, \quad \text{Var}(H) = \lambda.}$$

Thus the same parameter λ determines both the typical number of events and the size of the fluctuations:

$$\sigma_H = \sqrt{\lambda}.$$

The relative fluctuations therefore scale as

$$\frac{\sigma_H}{\mathbb{E}[H]} = \frac{1}{\sqrt{\lambda}}.$$

When is the Poisson model appropriate? The Poisson distribution is useful for counting events that are individually unlikely but have many opportunities to occur. Examples include

medicine: number of rare disease cases in a population,
 finance: number of defaults in a large portfolio over a short period,
 environment: number of extreme events observed in a fixed time interval.

In applications, Poisson models are often appropriate when events are rare, approximately independent, and occur at an approximately constant average rate.

2.3 Waiting for the First Success: The Geometric Distribution

The binomial distribution asks:

How many successes occur in a fixed number of trials?

A different question is:

How long must we wait until the first success?

Consider independent Bernoulli trials with success probability p . Let

T = number of trials until the first success.

For the first success to occur on trial k , the first $k - 1$ trials must fail and the k th trial must succeed:

$$\underbrace{F F \cdots F}_{k-1 \text{ failures}} S.$$

By independence,

$$P(T = k) = (1 - p)^{k-1} p.$$

Thus

$$\boxed{T \sim \text{Geometric}(p), \quad P(T = k) = (1 - p)^{k-1} p, \quad k = 1, 2, \dots}$$

Is the distribution normalized ? Using the geometric series,

$$\sum_{k=1}^{\infty} P(T = k) = p \sum_{k=1}^{\infty} (1 - p)^{k-1} = p \sum_{k=0}^{\infty} (1 - p)^k.$$

Using the identity

$$\sum_{k=0}^{\infty} q^k = \frac{1}{1 - q}, \quad |q| < 1,$$

we obtain

$$\sum_{k=1}^{\infty} P(T = k) = p \frac{1}{1 - (1 - p)} = 1.$$

Mean waiting time. The expected number of trials until the first success is

$$\mathbb{E}[T] = \frac{1}{p}$$

This result has a simple interpretation. If an event occurs with probability p on each independent trial, then we typically wait approximately $1/p$ trials to observe it. For example,

$$p = 0.1 \quad \implies \quad \mathbb{E}[T] = 10.$$

The Memoryless Property (One of the cool aspects of statistics)

The geometric distribution has a special property. Suppose we have already observed s failures. The probability that we must wait at least another t trials is

$$P(T > s + t \mid T > s).$$

Using conditional probability,

$$\begin{aligned} P(T > s + t \mid T > s) &= \frac{P(T > s + t)}{P(T > s)} \\ &= \frac{(1 - p)^{s+t}}{(1 - p)^s} \\ &= (1 - p)^t. \end{aligned}$$

But

$$P(T > t) = (1 - p)^t.$$

Therefore,

$$P(T > s + t \mid T > s) = P(T > t).$$

The process does not “remember” how long we have already waited. After every failure, the probability of success on the next trial remains p .

Summary

The three distributions considered above describe closely related questions:

Distribution	Question	Regime
Binomial(N, p)	How many successes in N trials?	N fixed
Poisson(λ)	How many rare events occur?	$N \rightarrow \infty, p \rightarrow 0, Np = \lambda$
Geometric(p)	How long until the first success?	repeat until success

For large N with p away from 0 and 1, the binomial distribution also approaches a Gaussian:

$$H \sim \text{Binomial}(N, p) \quad \Rightarrow \quad H \approx \mathcal{N}(Np, Np(1 - p)).$$

Thus a single elementary discrete model, repeated Bernoulli trials, already connects several of the most important distributions in probability.